

A validation of ADO drought indices SPI and SPEI in Slovenia

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1. Introduction

Drought indices are essential for tracking and warning of the possible drought related effects, impacts and outcomes (WMO and GWP, 2016). The use of a specific drought index in this manner requires a prior assessment of the index performance in detecting and monitoring different types of drought.

Agricultural drought affects plants and crops, sometimes resulting in extensive crop failure. As agricultural drought is particularly relevant for the Alpine region and the Slovenian case study area, we investigated the ability of SPI and SPEI to detect agricultural drought through validation against annual yield data for several crops and types of grassland, averaged to statistical regions (NUTS3 level). Prior to evaluating indices against yield data, we evaluated SPI and SPEI values calculated from ERA5 reanalyses against SPI and SPEI values calculated from ground observations at representative meteorological stations in Slovenia in order to learn whether the indices calculated in the Alpine domain are relevant on a national level.

2. Data

ERA5 reanalysis based indices in the Alpine domain

Two indices based on downscaled ERA5 reanalysis data were selected for validation against indices based on alternative ground observation data and annual yield data. The indices were chosen based on their characteristics and reported ability to identify and monitor agricultural drought; the Standardized Precipitation Index (SPI) and the Standardized Precipitation Evapotranspiration Index (SPEI), both on time scales of 2 and 3 months. They were calculated using normalised Gamma distribution for SPI and Generalised Logistic distribution for SPEI with limits set between -3 and 3 (Stagge et al., 2015). The calculation procedure closely follows the procedure from R package SCI with modification from monthly to daily time step.

Precipitation and potential evapotranspiration data used as input data for the calculation of the SPI and SPEI were downscaled from ERA5 reanalysis to 5 km resolution (Hersbach et al., 2018) using quantile mapping based on a UERRA reanalysis dataset (Bazile et al., 2017). Precipitation data was downscaled directly, whereas potential evapotranspiration was derived from other downscaled variables (temperature, wind speed, humidity, solar radiation) using the Penman-Monteith method. At the time of validation, the indices were available as gridded datasets for the period 1981-2018 with a daily update frequency.

Indices based on ground observation data in Slovenia

Indices based on ground observation data were calculated using meteorological data from 15 automatic meteorological stations in Slovenia, representative of the 15 regions within the framework of national drought monitoring (Figure 1). Precipitation is a measured variable, while reference evapotranspiration is a derived variable calculated daily using the Penman-Monteith equation. For the purposes of validation the variables were converted into climatological daily means: daily precipitation as an accumulation between 07 CET of the previous and 07 CET of the allocated day and daily reference evapotranspiration as an accumulation between 00 and 24 CET of the allocated day, derived from other meteorological variables. The indices were calculated using an identical approach as the ERA5 reanalysis based indices, with the key difference using point data rather than gridded data as input.



Figure 1: A map of 15 meteorological stations in Slovenia, representative of the 15 regions used in national drought monitoring

Yield data

Annual average yield data was obtained from the Statistical office of the Republic of Slovenia database for 4 different crops and 4 different grassland types (Table 1). Yield data is estimated for 12 statistical regions of Slovenia (NUTS3 level, Figure 2) and available from year 2000 onwards.

Table 1: A list of crops and grassland types with estimates of annual average yield

CROPS	GRASSLAND
Wheat and spelt	Grasses (including mixtures)
Grain maize and corn-cob mix	Grass - clover mixtures
Green maize	Clover - grass mixtures
Potatoes	Permanent grassland, including common grassland



Figure 2: A map of 12 statistical regions of Slovenia. Slovenian case study is located in the Podravska region.

3. Methodology

Validation of ADO indices against ground observation based indices

SPI and SPEI (-2 and -3) based on daily ground observation data from meteorological stations were compared with SPI and SPEI based on daily data of the corresponding downscaled ERA5 reanalysis grid-cells. Index values were compared qualitatively using timelines of monthly mean values and smoothed distributions of daily values and quantitatively using scatter plots with linear regression model and corresponding coefficient of determination r^2 for daily and monthly mean values respectively.

Validation of ADO indices against yield data

For validation against annual yield data, estimated on a regional level, we aggregated ADO SPI and SPEI values over the same 12 statistical regions. Mean monthly values, including the monthly spread of reanalysis based SPI and SPEI, were plotted together with annual yield data and linear regression analysis was performed to obtain the coefficient of determination (r^2) between annual yield data and index value averaged over the growing season between April and September for each year (values are indicated by red lines).

4. Results

Validation of ADO indices against ground observation based indices

Timelines of monthly mean values show strong agreement for both types of indices (Figures 3–4, example for SPI-2 and SPEI-2). Generally, dry and wet periods correspond, including peaks and lows of the timeline, yet there are some discrepancies in their magnitudes. Smoothed distributions of daily values also generally show a relatively strong agreement (Figures 5–6, example for SPI-2 and SPEI-2), but there appears to be greater agreement between SPI distributions than SPEI distributions. The largest differences between the two datasets appear in the shape of the distribution curve near the peak or around the mean values (between -1 and 1), thus in the normal condition range. Below -1 , where both SPI and SPEI indicate drought conditions, the distributions show very good agreement for the majority of the stations.

Scatter plots with linear regression further confirm the relatively strong agreement between the two datasets for both SPI and SPEI (Figures 7–10, example for SPI-2 and SPEI-2). Coefficient of determination (r^2) ranges from 0.62 to 0.79 and the agreement between the datasets is slightly larger for monthly mean values compared to daily values.

Timeline of monthly mean spi2, period 2000-2018

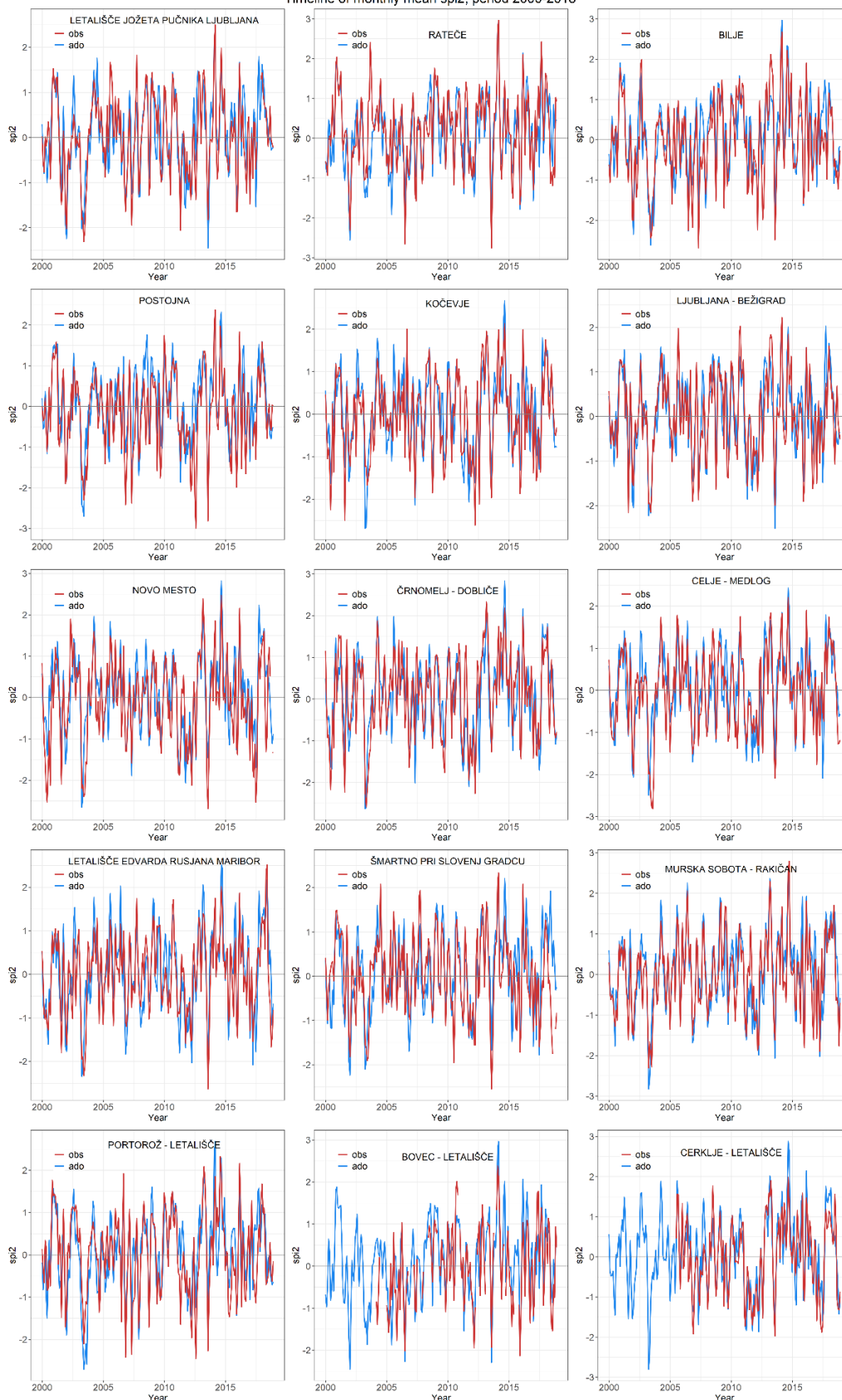


Figure 3: Timeline of monthly mean SPI-2 based on downscaled ERA5 reanalysis data (ado-blue) and ground observations (obs-red) for 15 meteorological stations in Slovenia

Timeline of monthly mean spei2, period 2000-2018

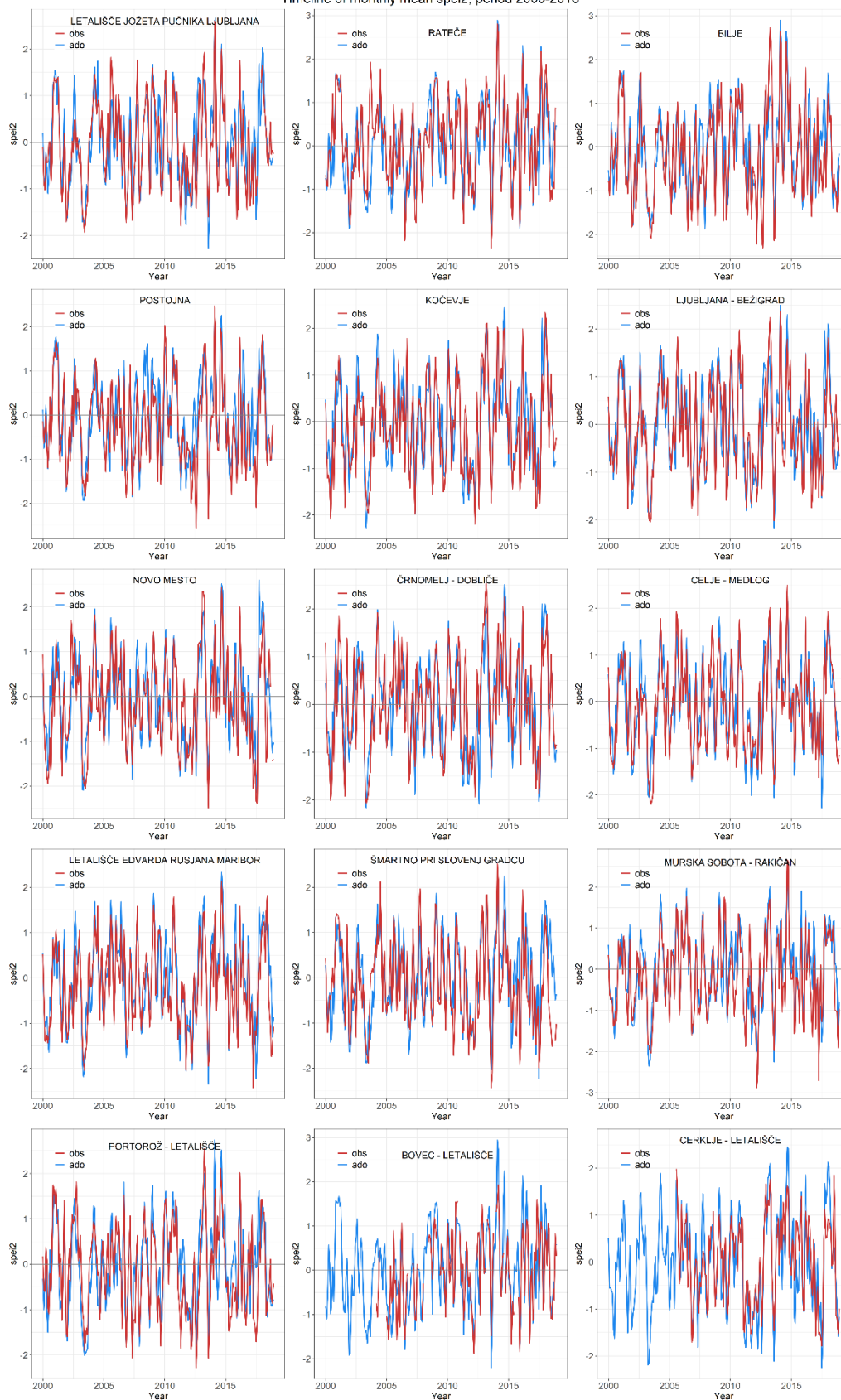


Figure 4: Timeline of monthly mean SPEI-2 based on downscaled ERA5 reanalysis data (ado-blue) and ground observations (obs-red) for 15 meteorological stations in Slovenia

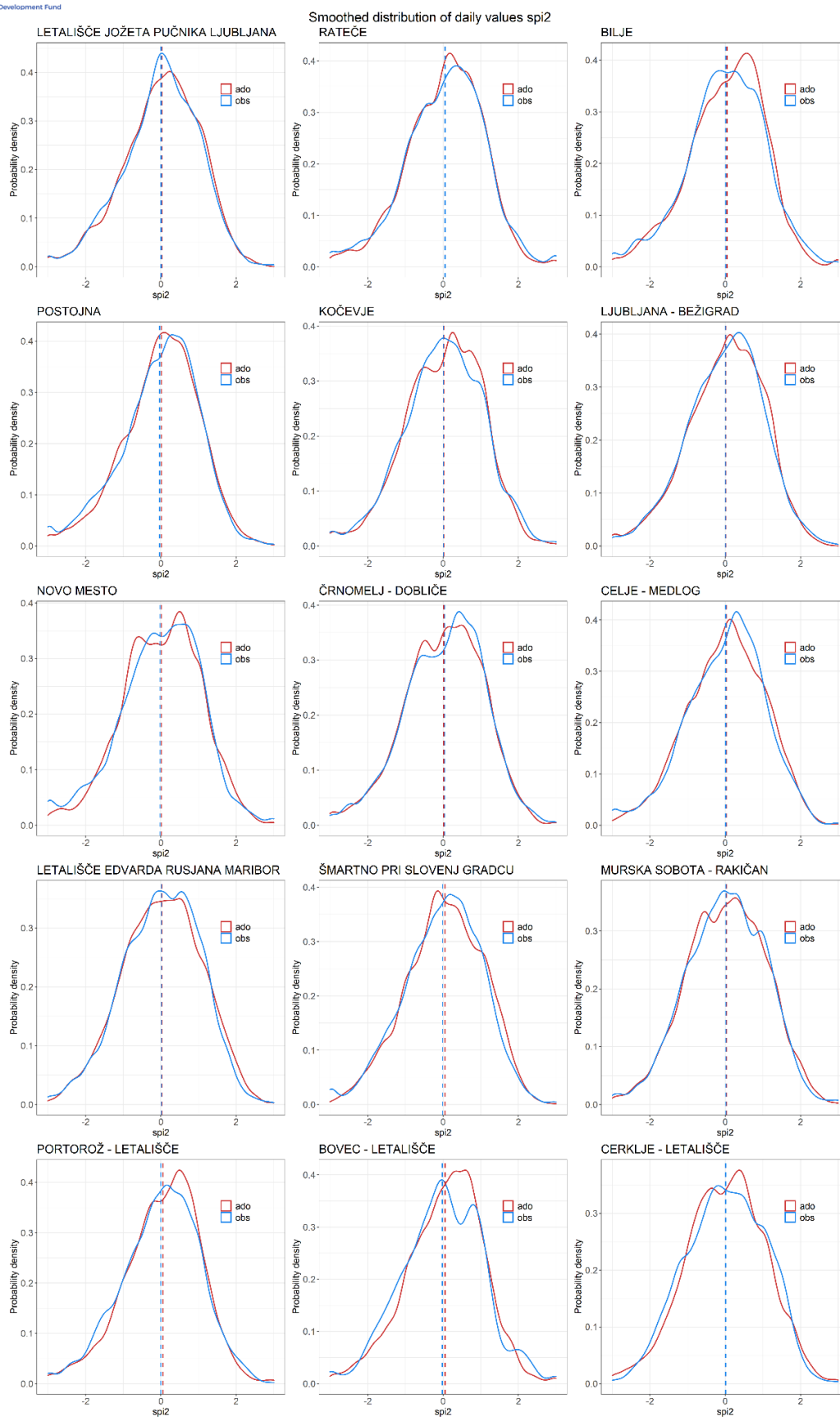


Figure 5: Smoothed distributions of daily SPI-2 values based on downscaled ERA5 reanalysis data (ado-blue) and ground observations (obs-red) for 15 meteorological stations in Slovenia

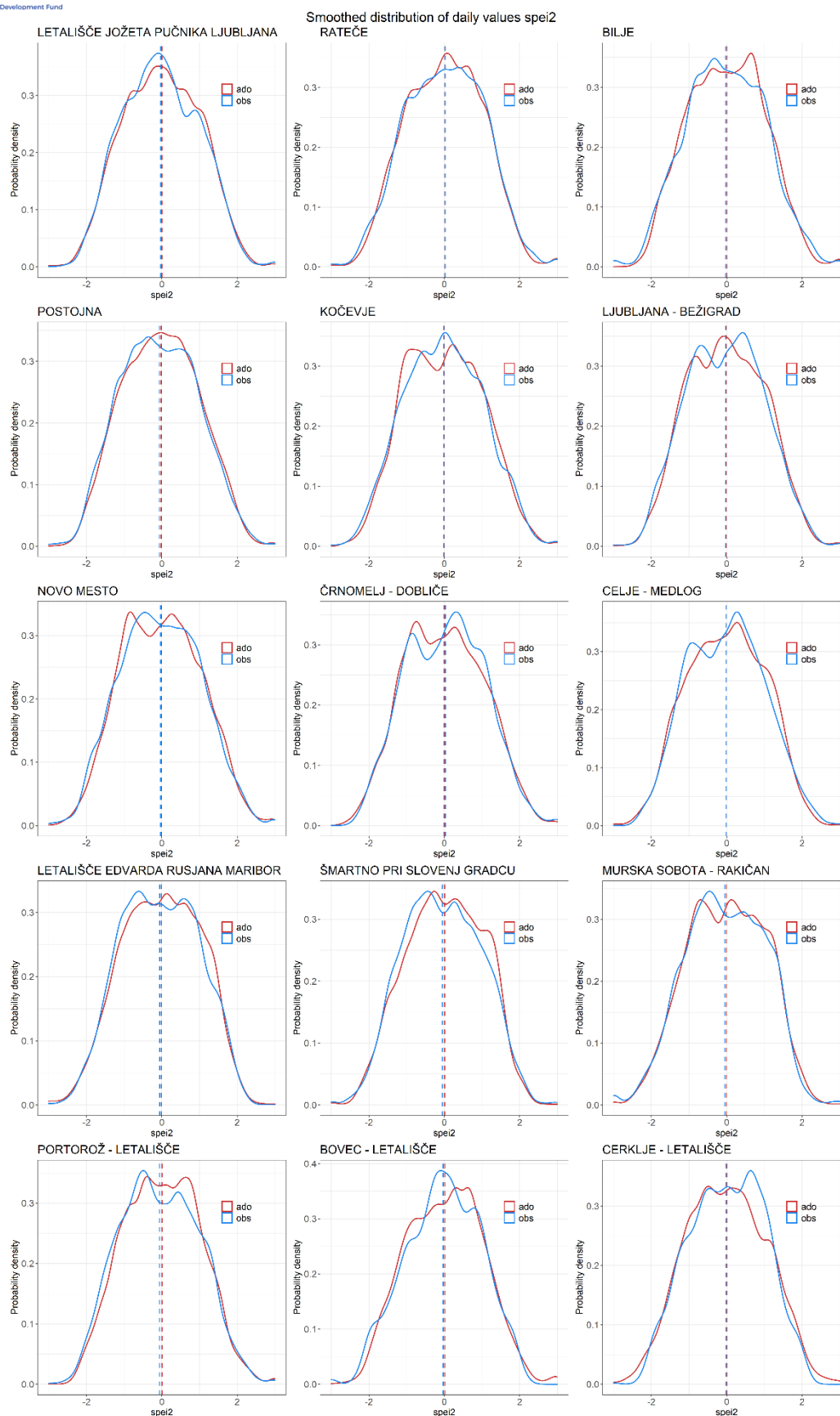


Figure 6: Smoothed distributions of daily SPEI-2 values based on downscaled ERA5 reanalysis data (ado-blue) and ground observations (obs-red) for 15 meteorological stations in Slovenia

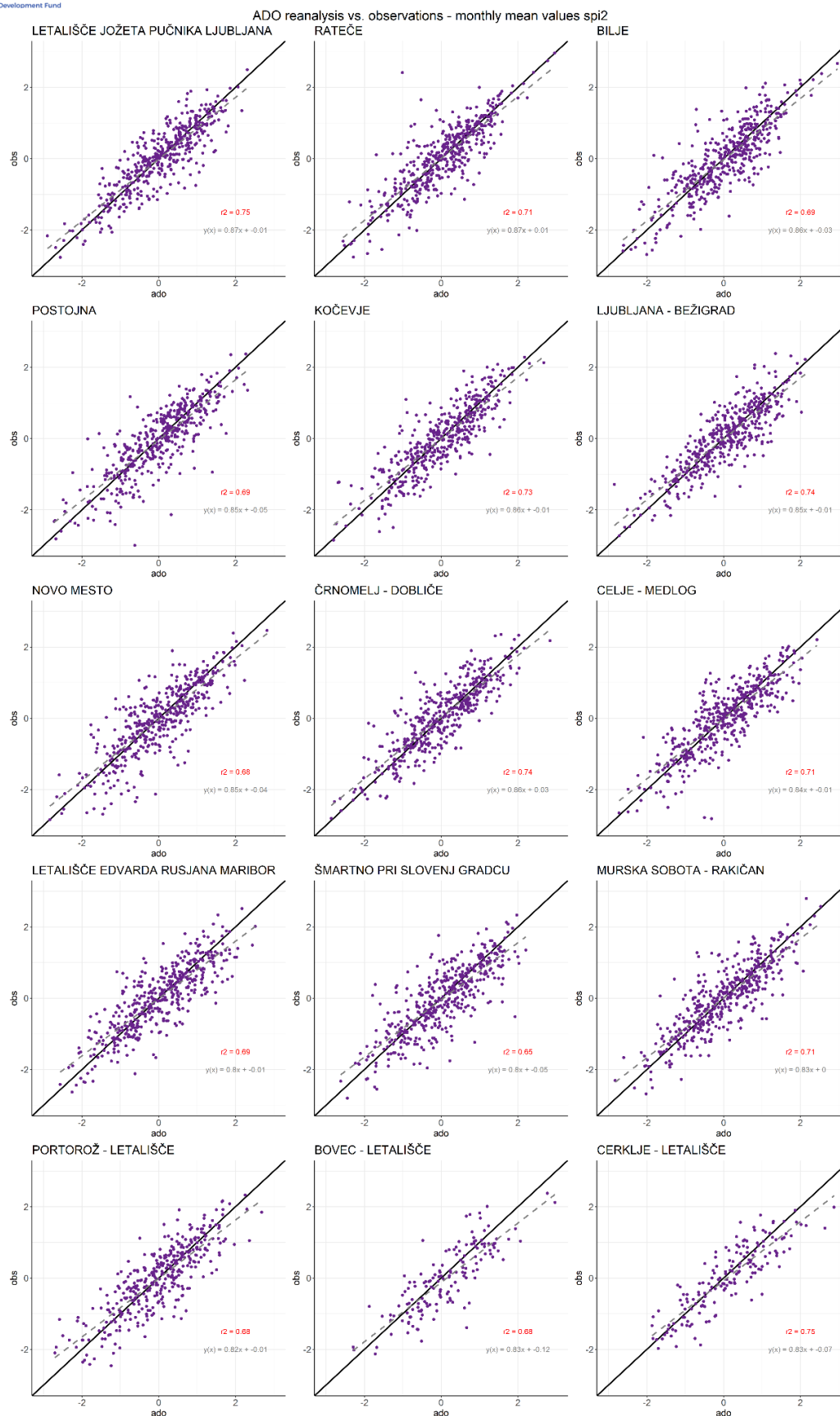


Figure 7: Monthly mean values of SPI-2 based on downscaled ERA5 reanalysis data vs. ground observations for 15 meteorological stations in Slovenia, with linear regression model and r^2

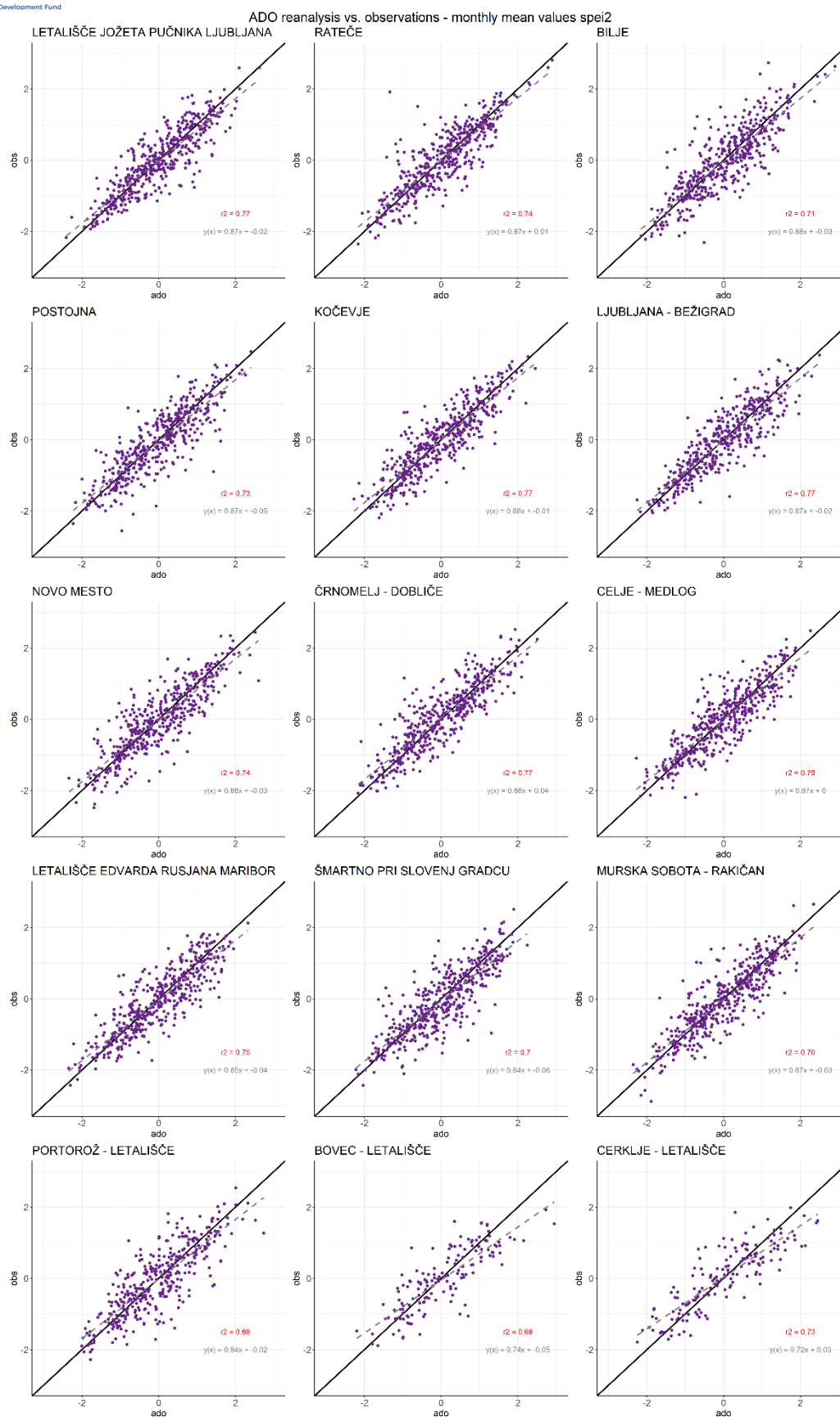


Figure 8: Monthly mean values of SPEI-2 based on downscaled ERA5 reanalysis data vs. ground observations for 15 meteorological stations in Slovenia, with linear regression model and r^2

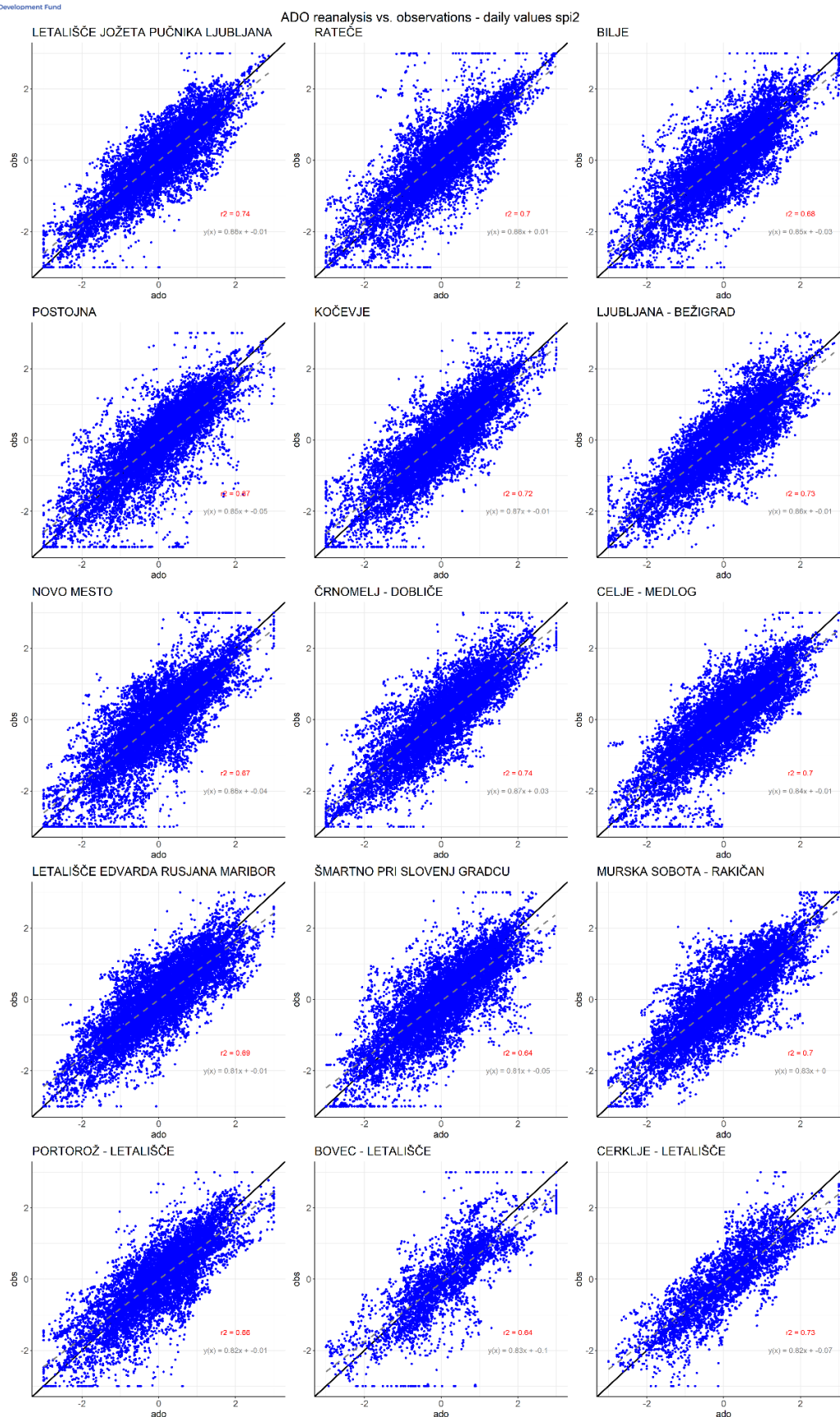


Figure 9: Daily values of SPI-2 based on downscaled ERA5 reanalysis data vs. ground observations for 15 meteorological stations in Slovenia, with linear regression model and r^2

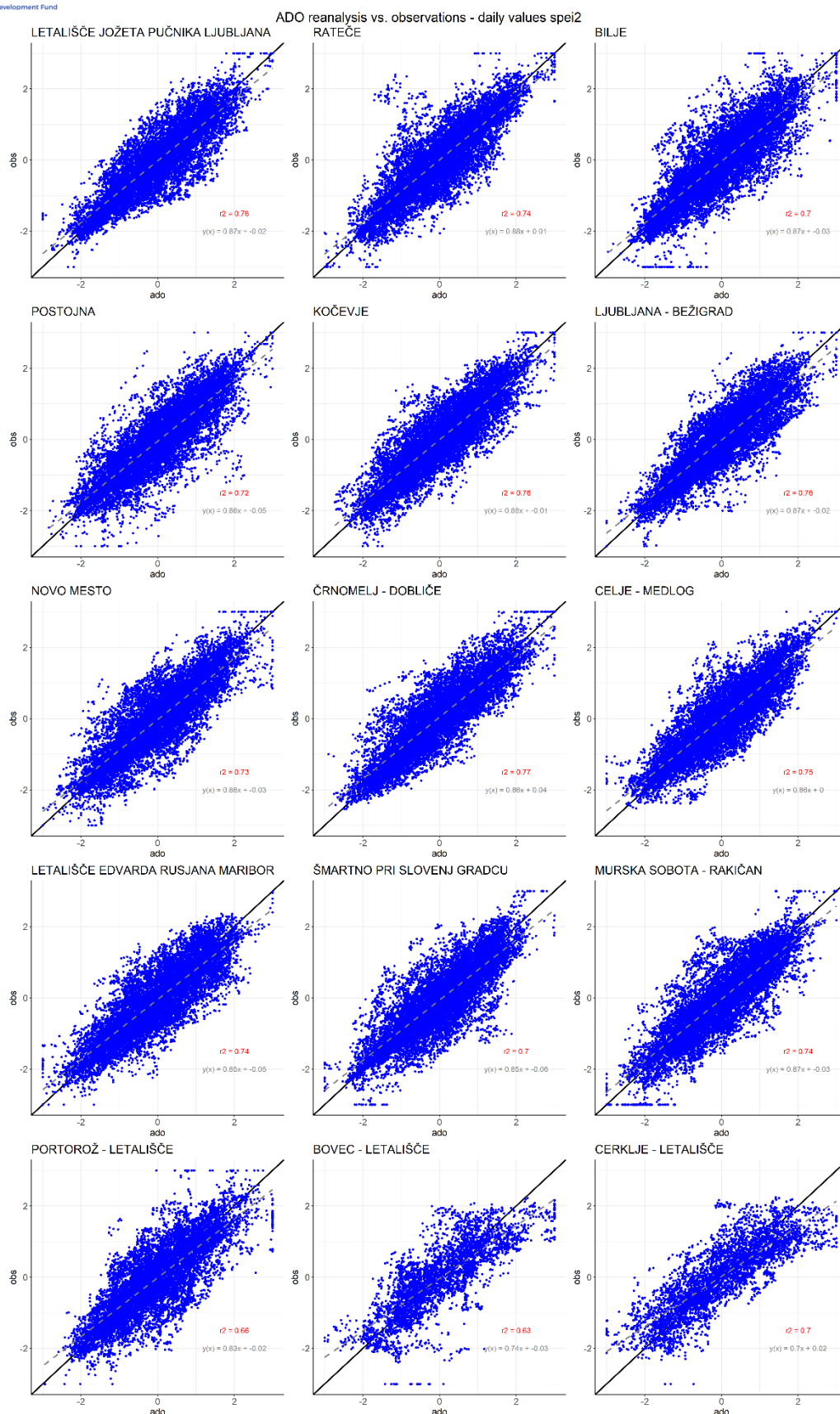


Figure 10: Daily values of SPEI-2 based on downscaled ERA5 reanalysis data vs. ground observations for 15 meteorological stations in Slovenia, with linear regression model and r^2

Validation against yield data

Annual yield data compared against SPI and SPEI values, averaged over growing season, shows different results for different regions and crop or grassland types. Generally, the best regression model results based on the coefficient of determination r^2 are found for regions in the east of Slovenia, namely Pomurska, Posavska, Podravska and Savinjska, with the latter two representing the two regions with the largest number of agricultural holdings in Slovenia (Ministry of Agriculture, Forestry and Food, 2020). In these regions, correlation is generally the highest for maize (either grain or green) among the crops and for grass mixtures and permanent grassland.

In Podravska region, where the Slovenian case study is located, the r^2 for SPEI-2 against grain maize and green maize is 0.31 and 0.18 respectively, while against grass-clover mixtures, clover-grass mixtures and permanent grassland it equals 0.21, 0.33 and 0.22 respectively (Figures 11–12). In terms of grassland, similar r^2 values can be observed for Pomurska and Savinjska regions, with the best linear regression model results found between SPEI-2 and permanent grassland in Pomurska ($r^2=0.35$) and for grasses (including mixtures) and permanent grassland in Savinjska ($r^2=0.37$ and $r=0.31$ respectively). In Posavska region, r^2 between SPEI-2 and grain maize and green maize equals 0.46 in both cases, whereas among the grassland the r^2 value is the highest for grasses (including mixtures) and grass-clover mixture (0.32 and 0.33 respectively).

Regression model results are similar for SPEI-3, with generally slightly smaller values of r^2 . For example, in the case study region Podravska, the r^2 for SPEI-3 against grain maize and green maize is 0.24 and 0.14 respectively, while for grass-clover mixtures, clover-grass mixtures and permanent grassland it equals 0.15, 0.25 and 0.14 respectively.

Similar pattern as in the case of SPEI can be observed for SPI-2 and SPI-3, with the best linear regression model results found for the four statistical regions in the east of Slovenia for maize (both grain and green) and all four types of grassland. Relatively high values of r^2 are also found in Gorenjska region for maize, potatoes and most grassland types (Figures 13–14).

Growing season averages of both SPI-2 and SPI-3 appear to correlate with grass yield in a similar manner. r^2 values for the four statistical regions in the east generally range between 0.15 and 0.38, with the highest r^2 value appearing for Savinjska region between SPI-2 and grasses (including mixtures) ($r^2=0.38$) (Figure 15).

Relatively high correlation (highest among the linear regression analysis results) also appears in Posavska region for green maize yield with r^2 at 0.49 for SPI-2 and 0.51 for SPI-3 (Figures 16–17), indicating that approximately 50 % of the variability in green maize yield is explained by SPI-2 and SPI-3 in this region.

Generally, higher r^2 values between annual yield data and growing season averaged index are observed for indices on a time scale of 2 months (SPI-2, SPEI-2) compared to 3 months (SPI-3, SPEI-3), indicating that 2-monthly indices might be more suitable in monitoring agricultural drought.

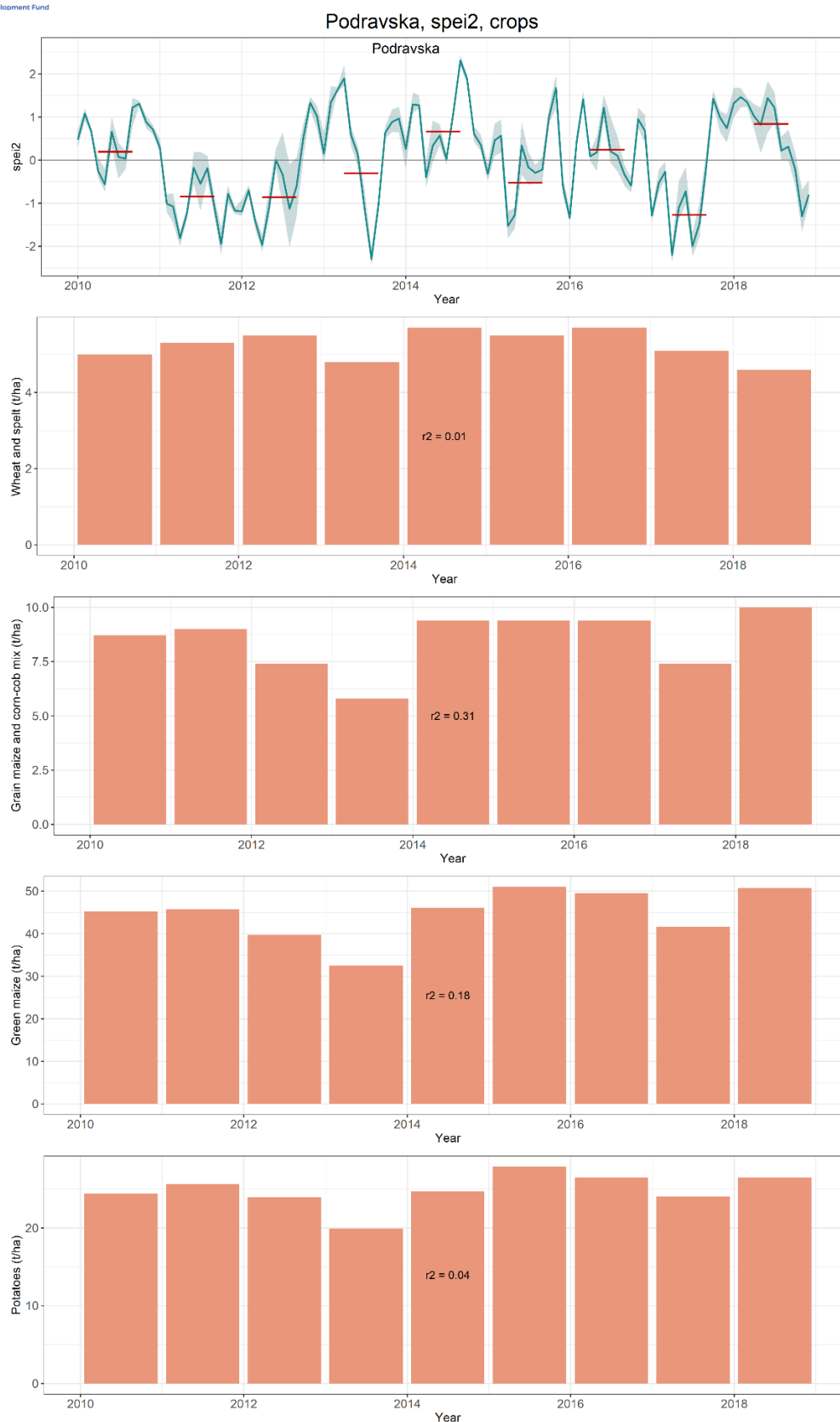


Figure 11: Above: Timeline of monthly mean SPEI-2 (including monthly spread) in Podravska region with SPEI-2 averaged over the growing season (Apr-Sep) indicated by red lines. Below: Annual mean yield for 4 different crops and the linear regression coefficient of determination r^2 between SPEI-2 averaged over the growing season and the annual yield data in Podravska region.

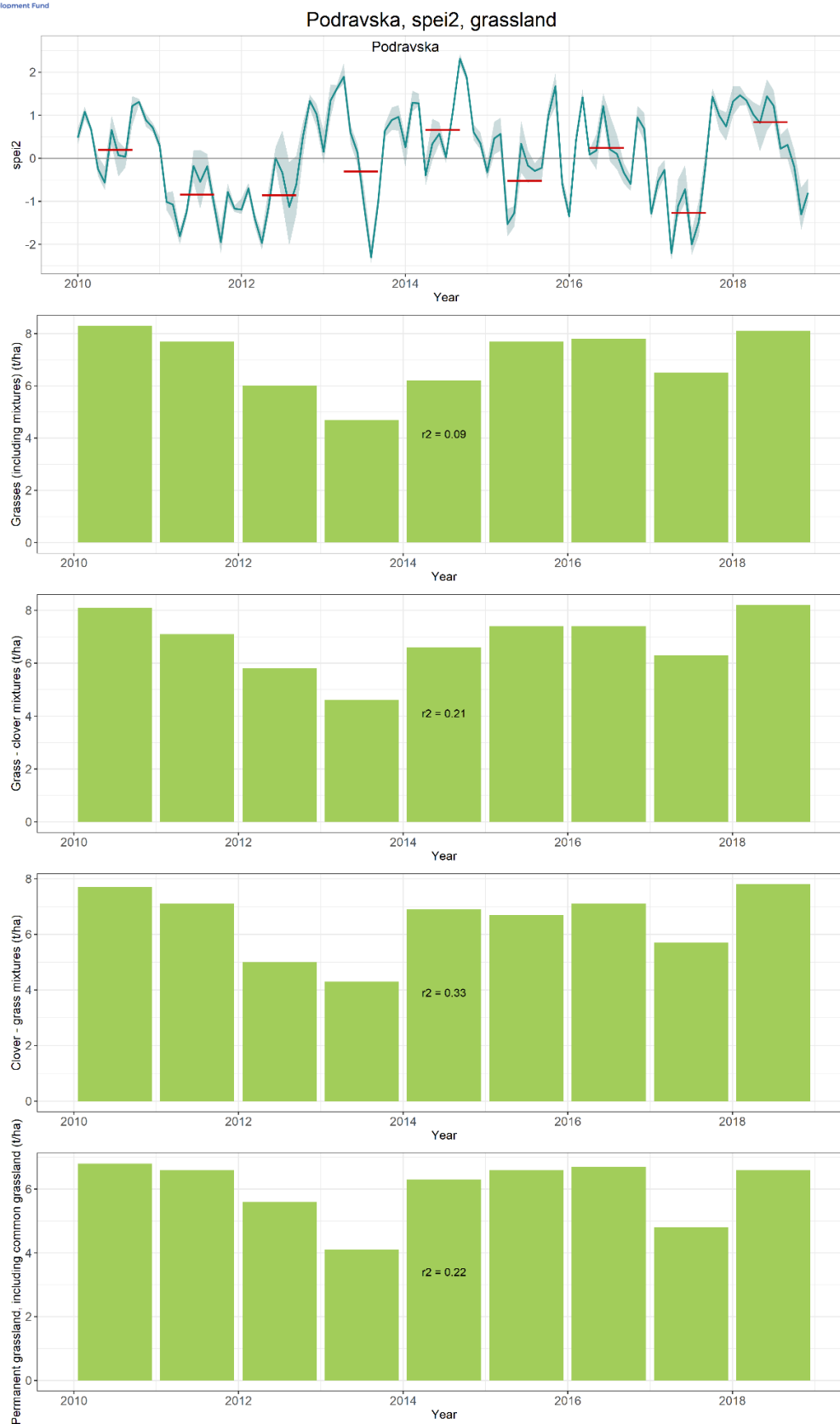


Figure 12: Above: Timeline of monthly mean SPEI-2 (including monthly spread) in Podravska region with SPEI-2 averaged over the growing season (Apr-Sep) indicated by red lines. Below: Annual mean yield for 4 different grassland types and the linear regression coefficient of determination r^2 between SPEI-2 averaged over the growing season and the annual yield data in Podravska region.

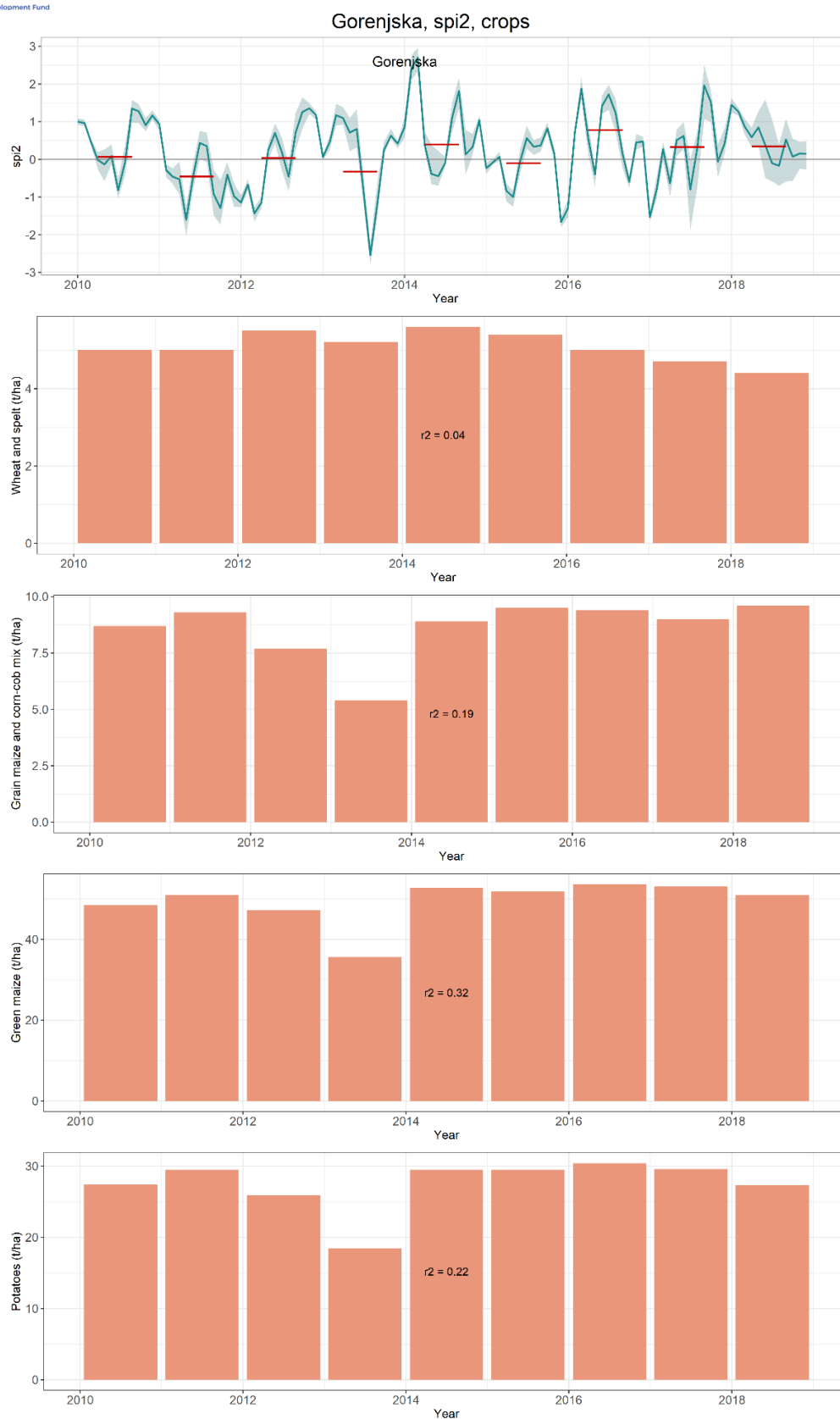


Figure 13: Above: Timeline of monthly mean SPI-2 (including monthly spread) in Gorenjska region with SPI-2 averaged over the growing season (Apr-Sep) indicated by red lines. Below: Annual mean yield for 4 different crops and the linear regression coefficient of determination r^2 between SPI-2 averaged over the growing season and the annual yield data in Gorenjska region.

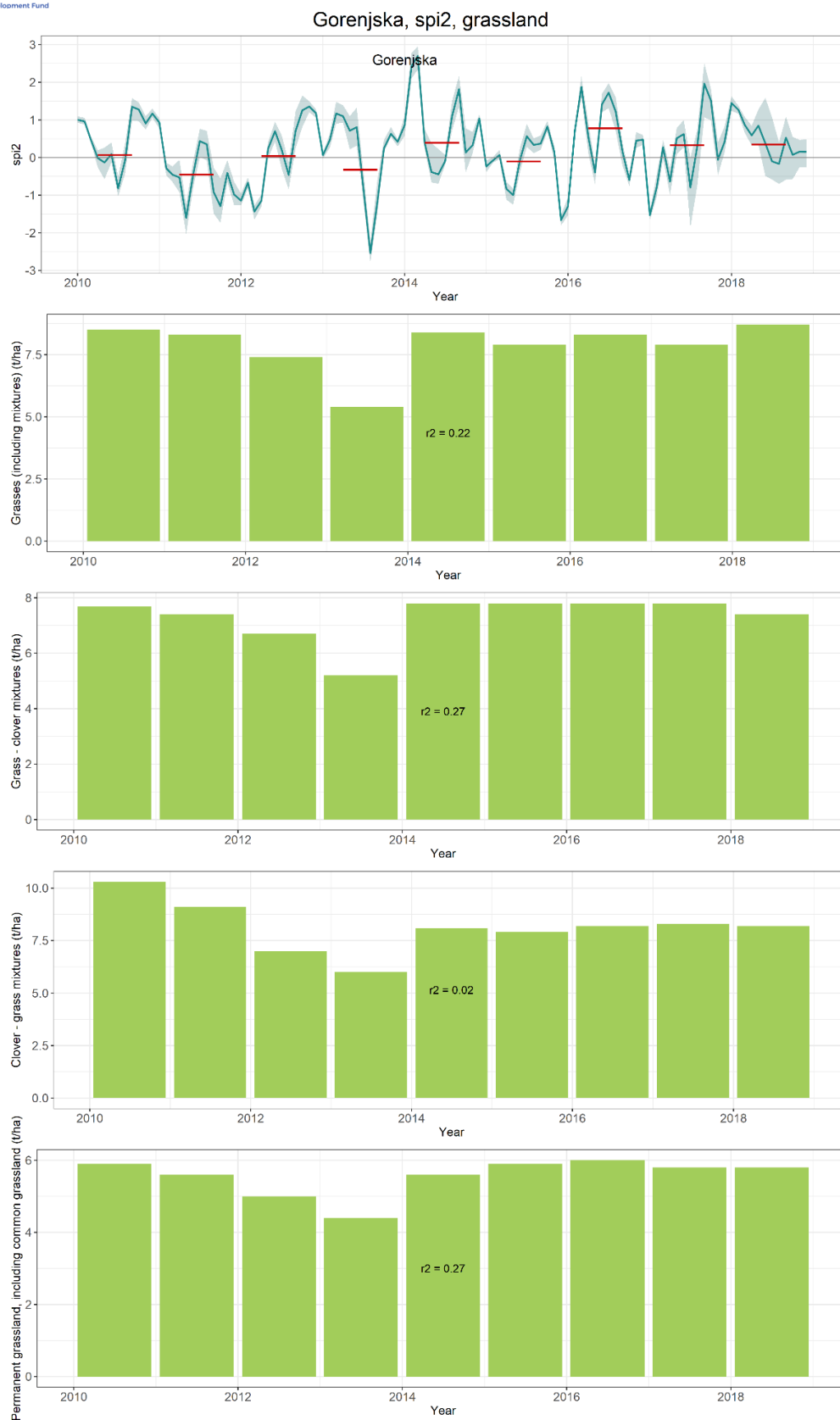


Figure 14: Above: Timeline of monthly mean SPI-2 (including monthly spread) in Gorenjska region with SPI-2 averaged over the growing season (Apr-Sep) indicated by red lines. Below: Annual mean yield for 4 different grassland types and the linear regression coefficient of determination r^2 between SPI-2 averaged over the growing season and the annual yield data in Gorenjska region.

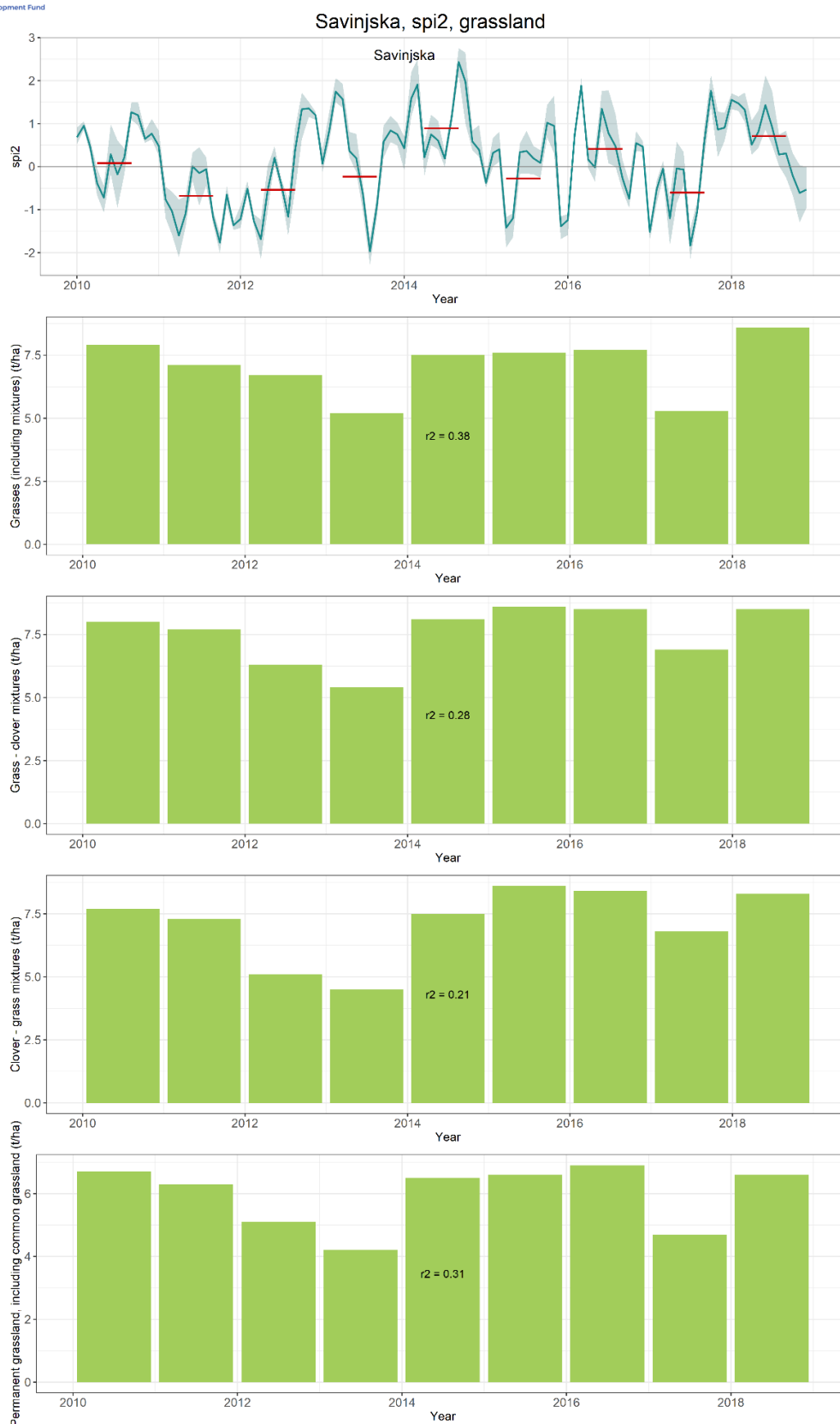


Figure 15: Above: Timeline of monthly mean SPI-2 (including monthly spread) in Savinjska region with SPI-2 averaged over the growing season (Apr-Sep) indicated by red lines. Below: Annual mean yield for 4 different grassland types and the linear regression coefficient of determination r^2 between SPI-2 averaged over the growing season and the annual yield data in Savinjska region.

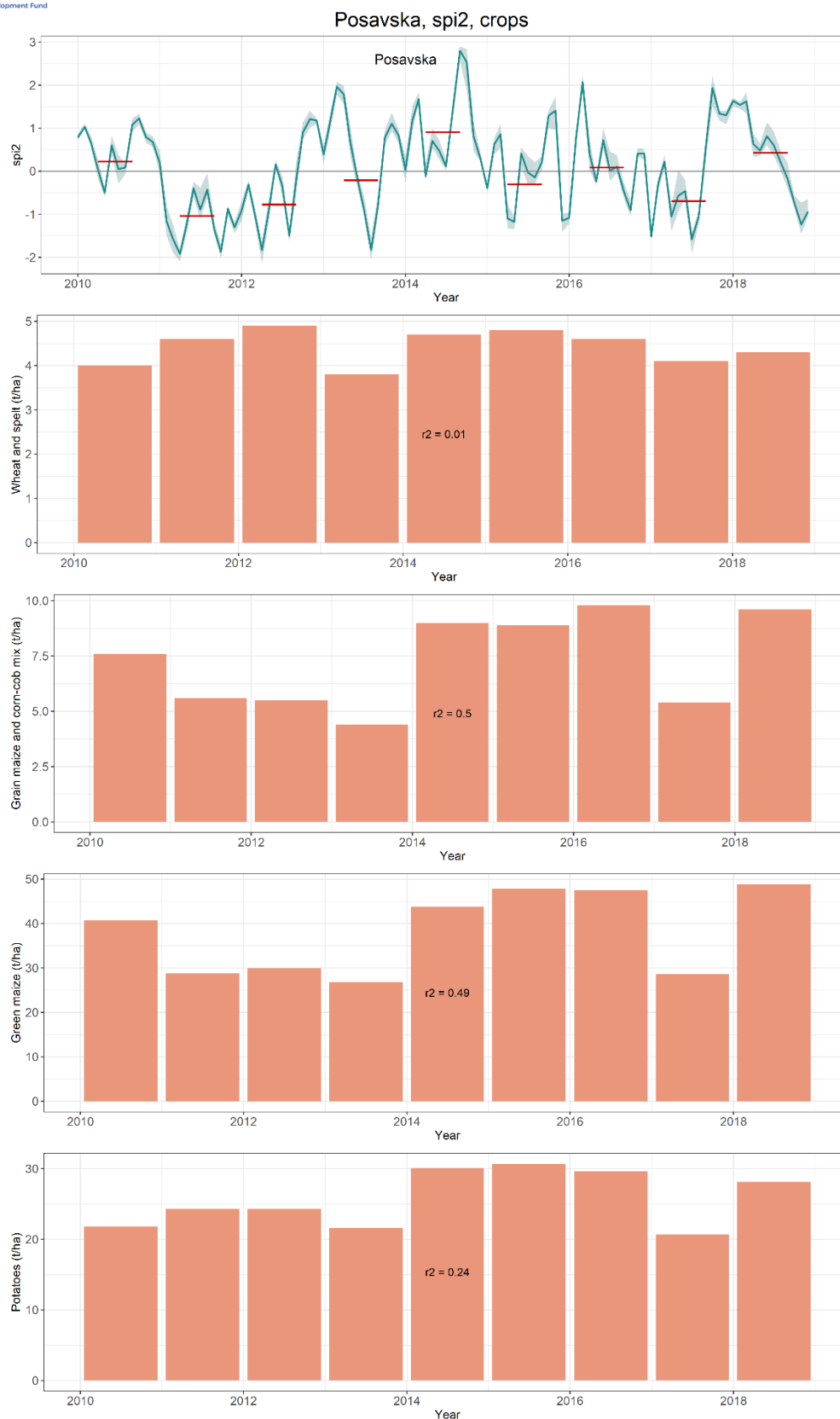


Figure 16: Above: Timeline of monthly mean SPI-2 (including monthly spread) in Posavska region with SPI-2 averaged over the growing season (Apr-Sep) indicated by red lines. Below: Annual mean yield for 4 different crops and the linear regression coefficient of determination r^2 between SPI-2 averaged over the growing season and the annual yield data in Posavska region.

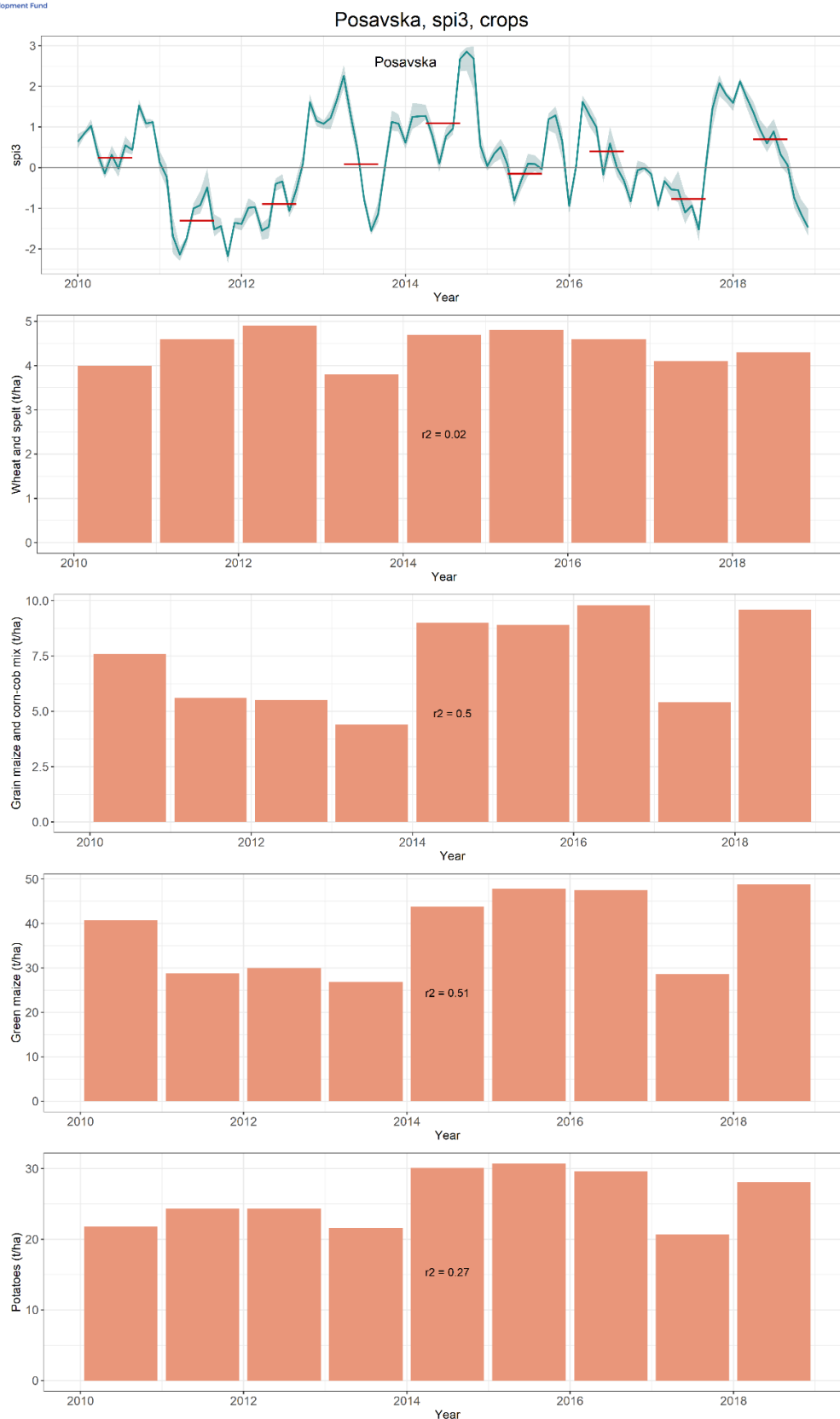


Figure 17: Above: Timeline of monthly mean SPI-3 (including monthly spread) in Posavska region with SPI-2 averaged over the growing season (Apr-Sep) indicated by red lines. Below: Annual mean yield for 4 different crops and the linear regression coefficient of determination r^2 between SPI-3 averaged over the growing season and the annual yield data in Posavska region.

5. Conclusions

The validation of ADO SPI and SPEI values calculated from downscaled ERA5 reanalysis against SPI and SPEI values calculated from ground observations at 15 representative meteorological stations in Slovenia has shown that the first set of indices calculated for the Alpine domain are relevant on a national level and suitable to use operationally. The linear regression model indicates high correlation, with values of r^2 ranging from 0.62 to 0.79 for both daily index values and monthly mean index values.

Validation of ADO SPI and SPEI values against annual yield data for different crops and grassland types has shown that agreement between the two is greater for regions in the east of Slovenia, including Podravska (case study area), and for maize (grain or green) and different grassland types, with linear regression based coefficient r^2 ranging from 0.15 to 0.51. The highest r^2 value at 0.51 is found between SPI-3 and green maize yield in Posavska region. In contrast to SPEI, SPI shows relatively good agreement with yield data in Gorenjska, which could indicate that in mountainous regions the evapotranspiration component in drought indices is less relevant. Generally, 2-monthly indices show better agreement with annual yield data compared to 3-monthly indices, indicating that 2-monthly indices might be more suitable in monitoring agricultural drought in Slovenia.

6. References

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